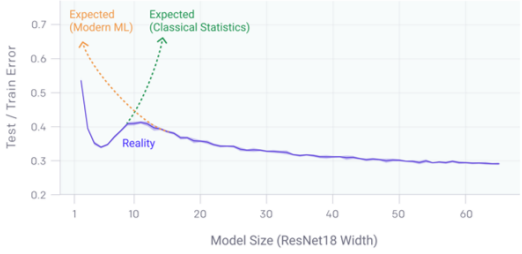
# Optimization for Deep Learning (OPT4DL)

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# Regularization

# Motivation

- We wish to optimize training loss to build models that generalize beyond the training set.
- Classical ML theory suggests reduction on generalization error tend to be as a result of regularization: constraining the set of values that our parameters might take (model capacity reduction)
- This view has let to certain approaches which we will view as noise injection
- However, in practice, our models have high capacity that can fit entire data (overparameterization and interpolation). And by doing so, despite classical ML prediction, generalize well too.

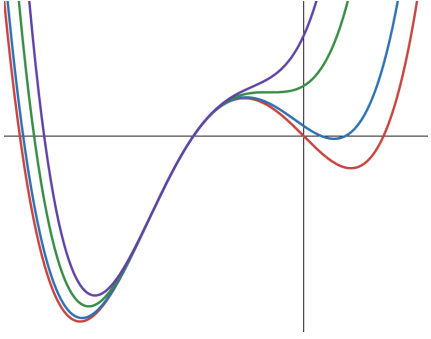


# Method A: $\ell$-2 Regularization

- One way to limit the model capacity is to constrain how large the parameters can get

$$\min_w \; f(w) + \lambda \|w\|_2^2, \quad f(w) = \mathbb{E}[\ell(w, z)] \tag{1}$$

- Leads to better landscape (getting rid of saddles, local max., sharp local min), good for both optimization and generalization. But we incur some bias







- Consider

$$\tilde{w}_j^* = \arg\min_w \; f(w) + \lambda_j \|w\|_2^2, \quad j = 1, 2 \tag{2}$$

- We can show

$$\|\tilde{w}_1^*\| \leq \|\tilde{w}_1^*\| \quad \Leftrightarrow \quad \lambda_1 \geq \lambda_2 \tag{3}$$

- Thus, as regularization becomes stronger optimal model becomes small and in the extreme case all weights will be zero (no learning capacity)
- Let $w^*$ be the solution of unregularized problem. Assuming a $L$-smooth loss $f$, we can lower bound the bias

$$\text{Bias} = \frac{\|w^* - \tilde{w}^*\|}{\|w^*\|} \geq \frac{\lambda_L}{1 + \lambda_L}, \quad \lambda_L = \lambda / L \tag{4}$$

- As $\lambda$ increases, the bias increases.




- Better landscape is good for optimization (easier functions). Finding higher quality solutions faster by (S)GD
- Consider GD for our regularized problem

$$\min_{w} \; f(w) + \lambda \|w\|_2^2, \quad f(w) = \mathbb{E}[\ell(w, z)] \tag{5}$$

We have

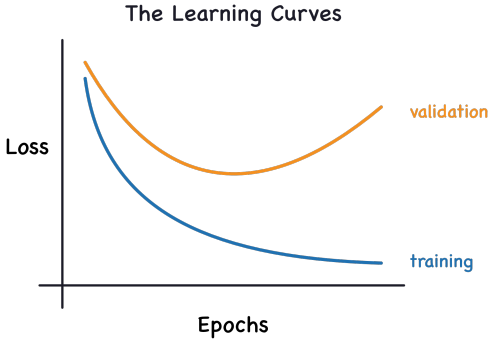
$$w_{w+1} = w_t - \eta(\nabla f(w_t) + \lambda w_t) = \underbrace{(1 - \eta\lambda)}_{\in(0,1)} w_t - \eta \nabla f(w_t) \tag{6}$$

called GD with Weight decay

- For simple methods like GD, $\ell$-2 Regularization and Weight decay are equivalent. But not true in general, e.g. ADAM

# Method B: Early Stopping

- When training, maintain a validation set and monitor the validation loss
- Continue training until validation loss starts to increase
- Then, stop earlier than what you originally had in mind
- By stopping early, we are restraining the model to reach its maximum learning capacity, aka regularization



# Connection Between $\ell$-2 Regularization and Early Stopping




- Consider a convex quadratic loss

$$f(w) = \frac{1}{2}(w - w^*)H(w - w^*), \quad w^* = \arg\min_w f(w) \tag{7}$$

and GD

$$w_{t+1} = w_t - \eta H(w_t - w^*), \quad w_1 = 0 \tag{8}$$

- Consider the Spectral decomposition $H = U\Lambda U^T$. Using simple steps we can show

$$\begin{aligned} U^\top w_t &= \Big(I_d(I_d - \eta\Lambda)^\top\Big) U^\top w_t \qquad \text{parameter at time } t \text{ of original prob.} \\ U^\top \tilde{w} &= \Big(I_d(\lambda I_d + \Lambda)^{-1}\lambda\Big) U^\top w_t \qquad \text{optimal solution of regularized problem} \end{aligned} \tag{9}$$

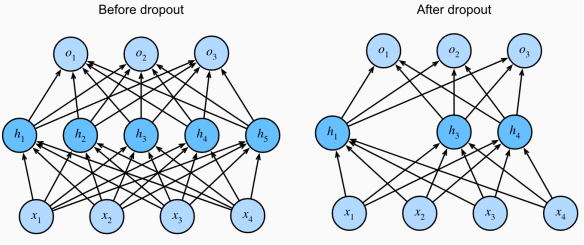
- $w_T \approx \tilde{w}$ if $\lambda T \approx \eta^{-1}$
- For a fixed learning rate, stopping early is like larger regularization.

# Method C: Dropout

- In training, dropout some neurons at random by zeroing out $p$ fraction of them

$$\tilde{h}_i = 0 \quad \text{with prob.} \quad p, \qquad \tilde{h}_i = \frac{h_i}{1-p} \quad \text{with prob.} \quad 1-p, \tag{10}$$

  such that $\mathbb{E}[\tilde{h}_i] = h_i$.
- By dropping neurons, we are restraining the model to reach its maximum learning capacity, aka regularization
- In testing, disabled (like batch normalization), unless for uncertainty quantification



# Connection Between $\ell$-2 Regularization and Dropout



- Dropout is essentially injecting noise into our model and hence the learning problem. The noise is independent of the activations and multiplicative in nature
- Let us use a simpler model to see the regularization effect of noise injection
- Consider a linear model

$$\min f(w) = \mathbb{E}_z[(y - \mathrm{w}^\top h)^2] \tag{11}$$

- Think of $h$ as the activation we will inject noise into. Here, for simplicity, an additive noise

$$h \to h + e, \quad e \text{ indep. of } x, \quad \mathbb{E}[e] = 0, \quad \mathbb{E}[ee^\top] = \Sigma \tag{12}$$

- Note $\Sigma = A^\top A \succeq 0$ is the covariance matrix and PSD
- We can show using simple steps

$$\mathbb{E}_{z,e}[(y - \mathrm{w}^\top (h + e))^2] = \mathbb{E}_z[(y - \mathrm{w}^\top h)^2] + \|Aw\|_2^2 \tag{13}$$

Note: A similar idea used in Gaussian Smoothing and ZO-SGD.

# Noise Injection and SAM




- We can also inject noise into $w$ directly. Also, we can consider a small worst case noise, as opposed to a random one

$$\min_w \mathbb{E}_{e,z}[\ell(w+e,z)], \qquad \min_w \max_{e:\ \|e\|\leq\rho} \mathbb{E}_z[\ell(w+e,z)] \tag{14}$$

- The second formulation, with some approximation leads to

$$\min_w \mathbb{E}_z[\ell(w,z)] + \rho\|\nabla\mathbb{E}_z[\ell(w,z)]\| \tag{15}$$

- Called sharpness-aware minimization (SAM), useful for improved generalization through penalizing sharpness/sensitivity
- We can penalize the growth of higher-order derivatives, e.g., Hessian, too.

# SAM and Improved Generalization

- Flatter Minima are likely to generalize better

